

Measuring the Dissociation Energy of Iodine by Absorption Spectroscopy

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The absorption spectrum for a sample of Iodine was obtained by passing white light through the sample into a grating spectrometer and photomultiplier tube. Using a Birge-Sponer extrapolation several parameters of the Morse potential were extracted from the data. The fundamental vibrational frequency of Iodine was found to be $1.0 \pm 0.1 \text{ cm}^{-1}$, while the degree of anharmonicity of the interatomic potential was found to be $130 \pm 10 \text{ cm}^{-1}$. From this the depth of the anharmonic potential well was determined to be $4300 \pm 600 \text{ cm}^{-1}$ and the dissociation energy was determined to be $4400 \pm 700 \text{ cm}^{-1}$. Both the well depth and dissociation energy lie within one standard deviation of their accepted literature values of 4398 cm^{-1} and 4335 cm^{-1} [1].

I. Introduction

A diatomic molecule such as Iodine has a number of electronic, vibrational and rotational energy levels. This experiment allows the parameters of the Morse potential to be determined. The electromagnetic spectrum of Iodine and its observed fine structure can be used to test the quantization of energy levels according to quantum mechanics. The Morse parameters found in this experiment can be used to find the equilibrium bond length of Iodine and predict thermodynamic properties of the element [2].

This experiment was first devised by Mansel Davies in 1951 [3] and was revised by Fred Stafford with newer equipment in 1962 [4]. Corrections to the numbering of the bands of the excited B electronic state with the vibrational quantum number v' were made in two papers published in 1965, by Brown and James [5] and by Steinfeld [6].

II. Theoretical background

A diatomic molecule can exist in various electronic states corresponding to different degrees of excitation of the electron cloud. A series of vibrational levels are associated with these excited states, each with its own rotational fine structure. According to the Boltzmann distribution law, when in thermal equilibrium, the majority of molecules are in the ground electronic state. Most of these molecules are in the lowest vibrational energy level of this ground state. When white light passes through a molecular vapour, if a photon has an energy that matches the energy difference between the ground state and a vibrational level in an upper electronic state, it will be absorbed. This leads to absorption bands in the spectrum of light passing through the vapour.

A. Morse Potential

The potential energy curve for the ground state of Iodine at small internuclear distances is well approximated by the Morse potential. The Schrodinger equation for a diatomic molecule in this potential can be solved exactly [7], with energy levels given by

$$\frac{E(v)}{hc} = G(v) = \omega_e(v + \frac{1}{2}) - \omega_e x_e(v + \frac{1}{2})^2 \quad v = 0, 1, 2, \dots \quad (1)$$

where ω_e is the fundamental vibration frequency, $\omega_e x_e$ is the degree of anharmonicity and v is the vibrational quantum number. A representation of the Morse potential for the ground electronic state of Iodine is shown in figure 1.

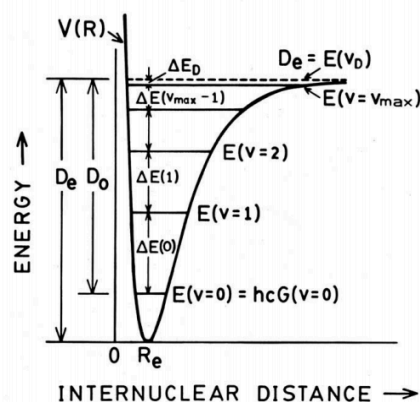


FIG. 1. A Morse potential diagram for an electronic state with associated vibrational energy levels [7].

The well depth depth of the potential D_e can be determined using the equation,

$$\frac{D_e}{hc} = \frac{\omega_e^2}{4\omega_e x_e} \quad (2)$$

while the zero-point energy is given by

$$\frac{E(v=0)}{hc} = \frac{\omega_e}{2} - \frac{\omega_e x_e}{4} \quad (3)$$

B. Birge-Sponer extrapolation

The energy difference between adjacent vibrational energy levels decreases as a linear function of the variable $v + 1/2$,

$$\Delta G(v + 1/2) = (\omega_e - \omega_e x_e) - 2\omega_e x_e (v + 1/2) \quad (4)$$

This equation defines a straight line with gradient $-2\omega_e x_e$ from which the anharmonicity parameter can be determined [7]. An example Birge Sponer plot is shown in figure 2. This corresponds to the Morse potential shown in figure 1.

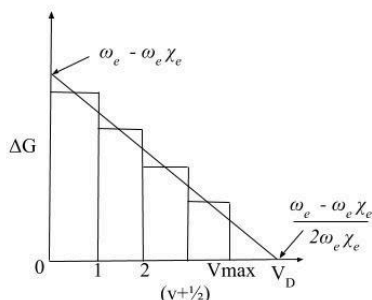


FIG. 2. A linear Birge-Sponer plot with difference in wavenumber against $v + 1/2$. The values of the x and y intercept in terms of ω_e and $\omega_e x_e$ are shown.

The area under the graph is given by

$$\text{Area} = \frac{1}{2} \left(\frac{\omega_e - \omega_e x_e}{2\omega_e x_e} \right) (\omega_e - \omega_e x_e) = \frac{(\omega_e - \omega_e x_e)^2}{4\omega_e x_e} = \frac{D_0}{hc} \quad (5)$$

From this equation the exact value of the dissociation energy can be determined.

III. Experimental method

An Iodine spectrum was obtained by passing white light from a car lamp through a collimator lens into a sample of molecular iodine. The light was then passed through a lens of focal length 75mm onto the entry slit of a spectrometer. The light was separated into different wavelengths by a blazed diffraction grating. The light was then reflected off a spherical mirror several times which collimated and focused it. The wavelength transmitted through the exit slit of the spectrometer was changed by rotating the diffraction grating, the amount of rotation was recorded by a micrometer gauge.

The width of the entry and exit slit of the spectrometer was adjusted to find a balance between the resolution of the spectrum, which improves as the slits are made narrower, and the digitisation of the spectrum due to the decrease in light flux. The optimum slit width was found to be around 0.8mm. The spectrum was converted into

digital form through the use of a photomultiplier tube, which detects the photocurrent from the exit slit. The experimental setup can be seen in figure 3.

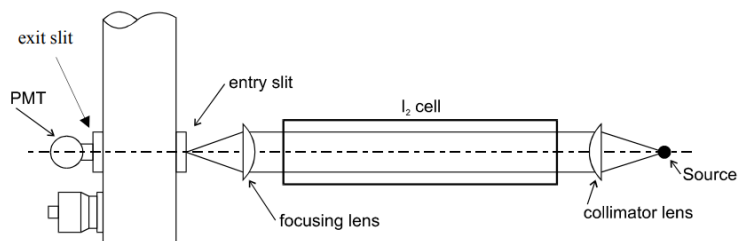


FIG. 3. Schematic diagram of the experimental setup [8]. The source used was a car lamp producing white light. A mercury source was positioned so as to also shine on the entry slit, allowing vibrational levels to be accurately numbered.

IV. Results

The wavelength of light which appeared at the exit slit of the spectrometer was found to be proportional to the reading on the micrometer gauge. A calibration was performed using the known wavelengths of the spectral absorption lines for a mercury source. The calibration data is shown in figure 4.

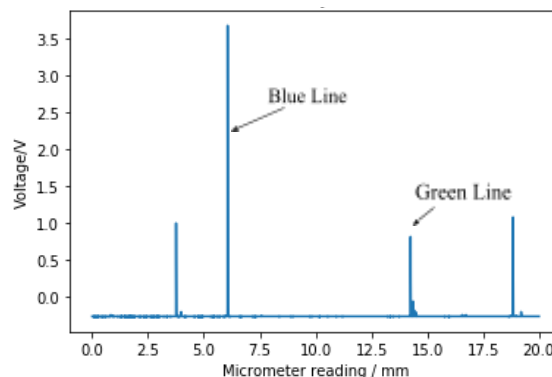


Figure 4. A plot of the spectral absorption lines of mercury. The lines used for the calibration included the strong blue line at 5.65 ± 0.01 mm and the green line at 14.08 ± 0.01 mm. The wavelengths for these lines were taken to be 435.835nm and 546.074nm respectively [8].

From this data the relationship between wavelength and micrometer reading was found to be linear with a small quadratic offset, the wavelength was given by the equation $\lambda = -0.0041x^2 + 13.16x + 361.2$ where x is the micrometer reading.

A spectrum for Iodine was taken from 10-20mm on the micrometer as this was found to be the area in which vibrational excitation occurred. The spectrum was obtained by setting the micrometer gauge to rotate at 0.2

revolutions per minute. This meant that the rotational fine structure of the spectrum was resolved. The spectrum is shown in figure 5.

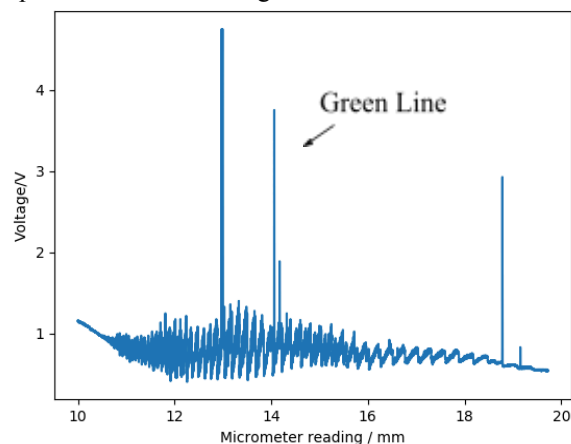


FIG. 5. A plot of the absorption spectrum for Iodine including the mercury spectral lines. The known wavenumber of the mercury green line was used to assign a quantum vibrational number of $v = 26$ to the nearest trough [2].

The absorption wavelengths were obtained using the calibration equation and the micrometer reading for each trough in the spectrum. A vibrational quantum number was assigned to each energy level. A Birge Spomer plot of the data is shown in figure 6.

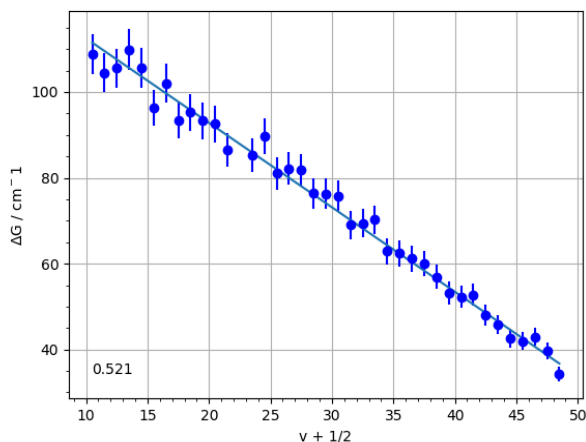


FIG. 6. A Birge-Spomer plot corresponding to the Iodine spectrum in figure 5. The chi-squared value for the graph is shown in the bottom left corner.

The equation of the line of best fit was found to be $y = (-1.97 \pm 0.06)x + (132 \pm 2)$. From this equation the values for the parameters of the Morse potential were obtained. The fundamental vibrational frequency was found to be $1.0 \pm 0.1 \text{ cm}^{-1}$, while the degree of anharmonicity of the interatomic potential was found to be $130 \pm 10 \text{ cm}^{-1}$. From these values the depth of the anharmonic potential well was determined to be

$4300 \pm 600 \text{ cm}^{-1}$ using equation. Using equation 5 the dissociation energy was determined to be $4400 \pm 700 \text{ cm}^{-1}$. The experiment was repeated with the iodine sample heated to 40 degrees celcius to measure the temperature dependence of the Morse potential. The depth of the anharmonic potential well was determined to be $4000 \pm 500 \text{ cm}^{-1}$ and the dissociation energy was determined to be $4000 \pm 700 \text{ cm}^{-1}$ at this temperature.

The errors on micrometer readings for each trough in the Iodine data were calculated by finding the maximum and minimum value that the trough could have and halving the difference between them. The level of uncertainty was affected by how digitised the spectrum was, this could be improved by increasing the slit width. However increasing the slit width led to a lower resolution of the spectrum, meaning the fine structure was unresolved. Other sources of error included variation in the temperature of the room and the rotational fine structure of the spectrum obscuring the true position of each vibrational trough.

V. Conclusion

The fundamental vibrational frequency, the anharmonicity constant, the well depth and dissociation energy for iodine vapour at room temperature were all found to be within one standard deviation of their accepted literature values. However this was due to a large uncertainty in the well depth and dissociation energies obtained. Digitisation was the main source of error in the experiment and could be reduced by using a light source of greater intensity. The temperature of Iodine was found to have little effect on the parameters of the Morse potential.

VI. References

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